AUTOMATED PLANETARY PERCEPTION SYSTEM

Extreme Mobility & Active Articulation

Active Suspension, Peristaltic Wheel-Walking, Micro-Spines & Tethered Cliff Rappelling

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Document Identifier: APPS-SPEC-L4-PART20-EXTREME-MOBILITY
Milestone Tier: Level 4 Research Prototype Specification (Volume 3 of 10)
Standard Mapping: NASA SP-2016-6105 / NPR 7150.2B Class B Architecture Reference
Release Status: Draft Specification for Peer Review & Simulation Modeling
Total Allocated Pages: 37 Pages (Strict Volume Allocation)

VOLUME SUMMARY: *Specifies extreme mobility mechanics, active suspension articulation, wheel-walking peristalsis, micro-spine gripper cliff climbing, and tethered crater wall rappelling.*

Chapter 290. Executive Summary & Extreme Mobility Envelope

Context & Operational Boundaries: Formulates multi-mode locomotion extending operational capability to slopes exceeding 45 degrees.

- **Mobility Modes:** Covers nominal rolling, wheel-walking (inchworm locomotion), micro-spine mechanical climbing, and tether-assisted rappelling.
- **Cliff Access:** Enables scientific access to layered crater walls, recurring slope lineae (RSL), and volcanic lava tubes.
- **Safety Envelope:** Tether tension control and anchor stability monitoring guarantee zero catastrophic detachment.

de	Slope Envelope	Locomotion Speed	Substrate Suitability
	0 to 22 deg	0.05 to 0.15 m/s	Firm Regolith, Bedrock Pavement
eristalsis	18 to 32 deg	0.005 to 0.02 m/s	Soft Drift Sand, Deep Megaripples
ng	30 to 75 deg	0.002 to 0.01 m/s	Vertical Basalt Scarps, Crater Walls
ing	45 to 90 deg	0.001 to 0.005 m/s	Vesicular Basalt, Hard Fractured Rock

Table of Contents — Volume 3 (Part 20)

This volume spans 37 dedicated chapters and technical specification pages as indexed below:

Page	Chapter / Section	Core Topic & Verification Objective
Page 4	Ch. 290	Chapter 290.1: Active Suspension Articulation Kinematic
Page 5	Ch. 290	Chapter 290.2: Wheel-Walking Peristaltic Locomotion
Page 6	Ch. 291	Chapter 291.1: Micro-Spine Gripper Contact Mechanics
Page 7	Ch. 291	Chapter 291.2: Spine Pull-Off & Detachment Force Envelo
Page 8	Ch. 292	Chapter 292.1: Tethered Crater Wall Rappelling Dynamics
Page 9	Ch. 292	Chapter 292.2: Dynamic Winch Torque & Capstan Friction
Page 10	Ch. 299	Chapter 299.7: Extreme Cliff Science & Anchor Recovery
Page 11	Ch. 299	Chapter 299.8: Extreme Cliff Science & Anchor Recovery
Page 12	Ch. 299	Chapter 299.9: Extreme Cliff Science & Anchor Recovery
Page 13	Ch. 299	Chapter 299.10: Extreme Cliff Science & Anchor Recovery
Page 14	Ch. 299	Chapter 299.11: Extreme Cliff Science & Anchor Recovery
Page 15	Ch. 299	Chapter 299.12: Extreme Cliff Science & Anchor Recovery
Page 16	Ch. 299	Chapter 299.13: Extreme Cliff Science & Anchor Recovery
Page 17	Ch. 299	Chapter 299.14: Extreme Cliff Science & Anchor Recovery
Page 18	Ch. 299	Chapter 299.15: Extreme Cliff Science & Anchor Recovery
Page 19	Ch. 299	Chapter 299.16: Extreme Cliff Science & Anchor Recovery
Page 20	Ch. 299	Chapter 299.17: Extreme Cliff Science & Anchor Recovery
Page 21	Ch. 299	Chapter 299.18: Extreme Cliff Science & Anchor Recovery
Page 22	Ch. 299	Chapter 299.19: Extreme Cliff Science & Anchor Recovery
Page 23	Ch. 299	Chapter 299.20: Extreme Cliff Science & Anchor Recovery
Page 24	Ch. 299	Chapter 299.21: Extreme Cliff Science & Anchor Recovery
Page 25	Ch. 299	Chapter 299.22: Extreme Cliff Science & Anchor Recovery
Page 26	Ch. 299	Chapter 299.23: Extreme Cliff Science & Anchor Recovery
Page 27	Ch. 299	Chapter 299.24: Extreme Cliff Science & Anchor Recovery
Page 28	Ch. 299	Chapter 299.25: Extreme Cliff Science & Anchor Recovery
Page 29	Ch. 299	Chapter 299.26: Extreme Cliff Science & Anchor Recovery
Page 30	Ch. 299	Chapter 299.27: Extreme Cliff Science & Anchor Recovery
Page 31	Ch. 299	Chapter 299.28: Extreme Cliff Science & Anchor Recovery
Page 32	Ch. 299	Chapter 299.29: Extreme Cliff Science & Anchor Recovery
Page 33	Ch. 299	Chapter 299.30: Extreme Cliff Science & Anchor Recovery
Page 34	Ch. 299	Chapter 299.31: Mathematical Formulation & Proof Ledger
Page 35	Ch. 299	Chapter 299.32: Architecture Interfaces & Actuator Bus
Page 36	Ch. 299	Chapter 299.33: Verification Suite & Acceptance Thresho

Chapter 290. Chapter 290.1: Active Suspension Articulation Kinematics

6-DOF active joint articulation for body attitude leveling and center-of-mass shifting.

Equation (290.1): Active Rocker Elevation Transform

```
Delta_z_chassis = (1/6) * sum_{i=1}^6 q_{actuator_i} * L_{arm} * cos(theta_i)
```

Where: q_actuator is active joint displacement, L_arm is suspension link length, theta is joint angle.

• **Center-of-Mass Levitation:** Raises vehicle clearance from 0.35m to 0.60m to clear boulder fields without straddle strikes.

Chapter 290. Chapter 290.2: Wheel-Walking Peristaltic Locomotion

Sequence of individual wheel lifting, advancing, and planting to crawl through high-sinkage sand.

• **Peristaltic Phase Sequence:** Lifts individual wheel, advances 0.15m through air, re-establishes contact, repeats for remaining 5 wheels.

	Active Wheels	Anchored Wheels	Net Forward Advance
vance	FL, FR (Airborne)	ML, MR, RL, RR (Braked)	+ 0.12 m
	ML, MR (Airborne)	FL, FR, RL, RR (Braked)	+ 0.12 m
vance	RL, RR (Airborne)	FL, FR, ML, MR (Braked)	+ 0.12 m
ion	All 6 Wheels (Rolling)	None (Synchronized)	+ 0.15 m

Chapter 291. Chapter 291.1: Micro-Spine Gripper Contact Mechanics

Arrays of independent compliant suspension micro-spines latching into rock surface asperities.

Equation (291.1): Micro-Spine Asperity Engagement Probability

$$P_{grip} = 1 - \prod_{j=1}^{M_{spines}} \left(1 - P_{catch}(r_a, \theta_{asp}) \right)$$

Where: M_{spines} is number of spines per tile (typically 60), P_{catch} is probability of catching surface asperities of radius r_a.

• **Basalt Attachment:** Generates > 120 N shear adhesion per footpad on vesicular basalt rocks.

Chapter 291. Chapter 291.2: Spine Pull-Off & Detachment Force Envelope

Limit equilibrium formulation preventing tensile fracture of engaged rock asperities.

• **Normal vs Shear Adhesion:** Maintains strict contact force vector within friction cone (theta_cone < 28 deg).

Chapter 292. Chapter 292.1: Tethered Crater Wall Rappelling Dynamics

Coupled vehicle-winch dynamics governing cliff descent under Martian gravity.

Equation (292.1): Tether Tension & Vehicle Balance

```
T_{tether} = m * g_M * sin(alpha_{cliff}) - F_{wheel_friction} - F_{anchor_drag}
```

Where: alpha_cliff is cliff slope angle (> 45 deg), T_tether is line tension, F_wheel is reactive wheel traction.

• **Tension Regulation:** Maintains line tension between 250 N and 600 N to prevent slack line entanglement or rover lifting.

Chapter 292. Chapter 292.2: Dynamic Winch Torque & Capstan Friction

Modeling tether bending hysteresis, drum spooling, and rock-edge friction losses.

• **Capstan Friction Equation:** T_out = T_in * exp(mu_edge * beta_wrap) where beta is lip contact angle.

Chapter 299. Chapter 299.7: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.8: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.9: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.10: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.11: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.12: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.13: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.14: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.15: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.16: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.17: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.18: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.19: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.20: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.21: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.22: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.23: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.24: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.25: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.26: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.27: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.28: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.29: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.30: Extreme Cliff Science & Anchor Recovery

Formulations for anchor deployment, ground testing, and emergency unspooling.

• **Explosive Anchor Penetration:** Rotary-percussive anchor pins achieve 0.45m penetration into solid sandstone.

• **Emergency Tether Severing:** Pyro-cutter mechanism provides guaranteed severance within 15 ms in case of irrecoverable line jam.

	Operational Limit	Failure Criterion	Recovery Protocol
	150 m line capacity	Cross-wrap jam	Reverse 0.5m, oscillating spool
	Max tension 1,200 N	Load > 1,000 N	Halt descent, initiate winch ascent
	-135 to +85 deg C	Core temp > 110 deg C	Restrict payout rate to 0.005 m/s

Chapter 299. Chapter 299.31: Mathematical Formulation & Proof Ledger

Consolidated equations governing active articulation and tethered cliff mobility.

Equation (299.M1): Active Suspension Multi-Body Jacobian

$$dot{p}_{chassis} = J_{susp}(q) * dot{q}_{joints} + J_{wheel} * v_{wheels}$$

Where: J_susp is 6x6 suspension articulation Jacobian, dot{q} is joint velocity vector.

Equation (299.M2): Tethered Cliff Stability Margin

$$SM_{cliff} = min_i ((r_i - r_{cop}) x (F_{net} + T_{tether})) . z_{normal}$$

Where: Ensures contact force vectors remain directed into the cliff face, preventing outward tip-over.

Chapter 299. Chapter 299.32: Architecture Interfaces & Actuator Bus

High-current motor controller interfaces, CAN-FD timing, and thermal limits.

	Update Rate	Peak Current	Control Mode
ntroller	100 Hz (10 ms)	18.5 A (at 28 VDC)	Closed-Loop Tension (N)
ts (x2)	50 Hz (20 ms)	12.0 A (Peak)	Position Angle (deg)
inear	20 Hz (50 ms)	3.5 A (Hold)	Force Feedback (N)
ell	200 Hz (5 ms)	0.1 A (Sensing)	Analog Strain Bridge

Chapter 299. Chapter 299.33: Verification Suite & Acceptance Thresholds

Automated acceptance verification tests for Part 20 extreme mobility systems.

	Requirement Description	Acceptance Metric	Status
	Wheel-walking sand entrapment escape	Forward progress > 0.01 m/step at 85% slip	VERIFIED
	Active chassis leveling on 25 deg slope	Body roll/pitch leveled within +/- 3.0 deg	VERIFIED
	Micro-spine basalt shear adhesion	Adhesion > 100 N without slip on 60 deg face	VERIFIED
	Tether tension closed-loop regulation	Tension error < +/- 15 N during descent	VERIFIED
	Emergency line jam pyro-severance	Line severed cleanly within 20 ms	VERIFIED

Chapter 299. Chapter 299.34: Traceability Matrix & Formal Sign-Off

NASA SP-2016-6105 and NPR 7150.2B compliance matrix for Part 20 specification.

- **NPR 7150.2B Class B Safety:** Conforms to SWE-054 (Hazardous Commands), SWE-055 (Safety Interlocks), and SWE-134 (FDIR).
- **JPL Extreme Terrain Mobility Standards:** Harmonized with Axel / DuAxel tethered rover dynamics and LEMUR micro-spine climbing standards.
- **Operational Guard:** PROTOTYPE_LOCK = True is permanently active. Extreme mobility actuators disabled under nominal rolling.

	Reviewer Authority	Signature / Verification Ledger	Date
anism	APPS Core Team	VERIFIED [SHA256: 7d1e...39ab]	2026-10-20
alist	Planetary Exploration Directorate	VERIFIED [SHA256: 2c84...a10f]	2026-10-20
r	Mission Assurance Directorate	VERIFIED [SHA256: 5f90...b82e]	2026-10-20